

Mouse auditory experiment

A first plan

October 8, 2026

We plan the sound setup for mice using our lab’s probes, with the aim of reusing the sound equipment for sedated sheep. SONIC tested awake sheep; our mouse state and tone settings remain open. (Methods 6.6, p.13)

1. Choose the sounds

- Choose the frequency, duration, loudness, and gap between tones.
- **SONIC:** initially explored 1.4–32 kHz; benchmark tones were 10 ms sine waves with 2 ms fades at each end. (Methods 6.1–6.2, p.11)
- **SONIC:** random frequencies, balanced within each block, with no silence between tones. (Methods 6.1, p.11)
- Save the sequence before playback, as SONIC did. (Methods 6.2, p.11)

2. Set up the speaker

- Choose sound equipment with later sheep use in mind; the appendix explains the proposed parts.
- **SONIC:** NI USB-6361 output at 192 kHz, TDT SA1 amplifier, and MF1 free-field speaker; about 80 dB SPL at sheep-ear distance. (Methods 6.2, p.11)
- **Our plan:** use the proposed shopping list on the next page, keep speaker placement fixed, and measure each tone with a calibrated microphone at the mouse-ear position.

3. Keep everything on one timeline

- Use our lab’s probes. Confirm the recording system’s timing input so it can record the tone markers alongside the neural signals.
- **SONIC:** recorded electrical sound output and trial-start markers; shared clock references aligned these with Connexus brain recordings. (Methods 6.2, p.11)
- **Our additions:** synchronize a microphone recording to check actual sound arrival. Align video frame times too if we add behavior video.

Reference: Perkins et al. (2025), SONIC: A Benchmarking Paradigm for Brain-Computer Interfaces, bioRxiv v1. Page numbers above refer to the printed manuscript. Mouse choices and additions are proposals.

Sound setup: shopping list

Use our lab's probes

USD

Budget for one of each item or group below. **Est.** means our rough allowance, not a confirmed supplier price.

What we get	What it does	Budget
1 TDT MF2-M speaker kit	Plays the tones.	\$1,500 est.
1 TDT SA1 amplifier	Drives the speaker.	\$2,000 est.
1 TDT ZB1PS power unit	Powers the amplifier.	\$750 est.
1 NI USB-6361	Tone output and timing.	\$3,030
1 complete Avisoft 116Hm kit	Records sound at the ear position.	\$8,160
Calibration support	Checks the sound levels.	\$1,500 est.
1 computer, monitor, and storage set	Runs playback and saves data.	\$2,000 est.
Cables, timing connections, and mounts	Connects equipment and fixes its position.	\$1,000 est.
Sound setup subtotal	Rounded budget	\$20,000

What is already included?

- The Avisoft kit includes the microphone, preamp, sound recorder, cable, stand, and software. Buy **one kit**.
- The MF2-M kit includes a stand and connection accessories. No extra speaker or microphone is needed for this budget.

What do we already have?

- Our lab provides the probes. Neural hardware and animal preparation are outside this sound budget.
- Reuse a suitable lab computer and storage if available; then remove the \$2,000 allowance above.

Allow for extras

- Keep **\$2,500 reserve** for price changes, shipping, and tax: sound budget **about \$22,500**.
- Optional behavior camera with accessories: **about \$1,100 extra**.

Before buying: Confirm the lab recorder's timing input. The microphone's stated band starts at 2 kHz; lower tones need a different choice. Calibration and timing need checking. A soundproof box is not required.

Sources, October 8, 2026: TDT kit, amplifier, power, NI, Avisoft kit, prices. SONIC used MF1; MF2 is our current candidate. Estimates/alternatives are detailed in the repository. We aim to reuse the sound hardware for Rice's sedated-sheep measurements after new calibration and timing checks.

Appendix: why these parts?

Mouse setup first; intended reuse for sheep

Why the speaker, amplifier, and power unit?

- **MF2-M speaker:** one speaker plays tones through the air. TDT lists a 1–80 kHz range, including SONIC’s 1.4–32 kHz band. This broad range is useful for a shared mouse/sheep setup. Actual output still needs measuring.
- **SA1 amplifier:** turns the NI waveform into power that drives the speaker. TDT explicitly supports SA1 with MF2; SONIC also used SA1.
- **ZB1PS power unit:** SA1 needs TDT’s zBus power. This chassis supplies it and houses the amplifier.

What SONIC used, and what we propose

Part	SONIC reported	Our proposal
Speaker	TDT MF1, through-air use	TDT MF2-M, current model
Amplifier	TDT SA1, 0 dB attenuation	TDT SA1
Power unit	Exact model not named	ZB1PS, per TDT’s requirement
Tone output	NI USB-6361, 192 kHz	NI USB-6361 for tones/markers
Sound-check mic	Not named in the paper	Avisoft 116Hm complete kit
Clock reference	TimeMachines TM2500C; PTP and 10 MHz	Confirm lab clock/timing inputs

SONIC Methods 6.2, p.11. Its reported sheep level was about 80 dB SPL; this is not our selected mouse level. The GPS server is a paper reference, not an added purchase.

What changes when we move to sheep?

- **Intended reuse:** speaker, amplifier, power unit, NI, computer, and sound recorder. We use our lab’s probes.
- **Set up and measure again:** change mounts, distance, and angle; calibrate each tone at the sheep-ear position and recheck sound arrival and recorder timing.
- **Check the microphone first:** the kit’s stated range starts at 2 kHz. SONIC’s 1.4 kHz tones, or other lower sheep tones, need a suitable alternative microphone.
- **Choose settings separately:** our sheep will be sedated; SONIC’s were awake. Reusing equipment does not establish the same brain responses or response windows.

Sources: SONIC v1, Methods 6.2 (p.11), 6.6 (p.13); MF2 range/compatibility, SA1 power, ZB1PS, microphone range. Transfer is a design aim; it has not been demonstrated with our equipment.